

2c. Content of Pure Core (Mandatory papers Y540 and Y541)

Introduction to Pure Core.

In **Pure Core** learners will extend and deepen their knowledge of proof, algebra, functions, calculus, vectors and differential equations studied in A Level Mathematics. They will also broaden their knowledge into other areas of pure mathematics that underpin the further study of mathematics and other numerate subjects with complex numbers, matrices, polar coordinates and hyperbolic functions.

4.06 Series

Simple series are summed, using standard formulae and the method of differences. Partial fractions are used to sum series.

4.07 Hyperbolic functions

Hyperbolic functions are introduced and their inverses are used in integration.

4.08 Further Calculus

Maclaurin series are used to approximate functions.